

The group $G(\mathfrak{m})$ is perfect: an exposition of a machine-checked proof

Abstract

We give a self-contained account of a formal proof that the group $G(\mathfrak{m})$ is perfect, i.e. equals its commutator subgroup (Theorem 3.3, label `thm: gsimple g m perfect`). $G(\mathfrak{m})$ is defined by generators and relations attached to the coefficients $c(j)$ of the modular function J . In the formalization $c: \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary parameter, and the only hypothesis is $c(2) \geq 1$. The proof shows that each generator lies in the commutator subgroup, in this order: the real-root unipotents $X_{-1}(u)$ and $Y_{-1}(u)$, then the imaginary-root unipotents $X_{\ell,jk}(u)$ and $Y_{\ell,jk}(u)$, then $H_1(s)$, and finally $H_2(s)$. Every lemma and the main theorem are machine-verified.

Source	Carbone et al., G-Simple (draft), <code>GSimple_Theorem_3_3</code>
Status	proof machine-verified (Lean 4, Mathlib)
Lemmas	26
Text	written by claude-opus-5-5, checked against the Lean source
Run	<code>srv_20261004_123235_GSimple_Theorem_3_3</code>

Contents

1	Setting	1
2	Lemmas	3
2.1	Generalities on the presentation	3
2.2	Unipotent generators of the real root and the elements $\tilde{w}$	4
2.3	Unipotent generators of imaginary roots	4
2.4	The torus generators	6
3	Proof of the theorem	7
A	Correspondence with the formal proof	8

1 Setting

Parameters and index set. Fix a function $c: \mathbb{N} \rightarrow \mathbb{N}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. The intended c is the coefficient function of $J(q)$, restricted to $j \geq 0$. Put

$$E = E_c = \{(\ell, j, k) \in \mathbb{N}^3 : \ell < j, 1 \leq k \leq c(j)\}.$$

Note that $(\ell, j, k) \in E$ forces $j \geq 1$ and $j - \ell \geq 1$. For $\ell, j \in \mathbb{N}$ with $\ell < j$ put

$$c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

Since $0 \leq \ell \leq j-1$, every factor is nonzero, so $c_{\ell j} \neq 0$ whenever $(\ell, j, k) \in E$. All divisions by $c_{\ell j}$ below are therefore genuine.

Generators. Let $\mathcal{X}$ be the set of symbols

$$H_1(s), H_2(s) \ (s \in \mathbb{C}^\times), \quad X_{-1}(u), Y_{-1}(u) \ (u \in \mathbb{C}), \quad X_{\ell,jk}(u), Y_{\ell,jk}(u) \ ((\ell, j, k) \in E, \ u \in \mathbb{C}).$$

Let $F = F(\mathcal{X})$ be the free group on $\mathcal{X}$. The commutator is $(a, b) = aba^{-1}b^{-1}$. In F we define the abbreviations

$$\tilde{w}_{-1}(s) = X_{-1}(s)Y_{-1}(-s^{-1})X_{-1}(s), \quad \tilde{w}_{\ell,jk}(s) = X_{\ell,jk}(s)Y_{\ell,jk}\left(\frac{-s^{-1}}{c_{\ell j}}\right)X_{\ell,jk}(s),$$

for $s \in \mathbb{C}^\times$ and $(\ell, j, k) \in E$. We also set $\tilde{w}_{-1} = \tilde{w}_{-1}(1)$ and $\tilde{w}_{\ell,jk} = \tilde{w}_{\ell,jk}(1)$.

Relations. The following equations are imposed for all $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$ and all indicated triples in E . An equation $a = b$ is imposed through the relator ab^{-1} . An equation $(a, b) = 1$ is imposed through the relator (a, b) .

$$H_1(s)H_1(t) = H_1(st), \tag{H:1a}$$

$$H_2(s)H_2(t) = H_2(st), \tag{H:1b}$$

$$H_1(s)H_2(t) = H_2(t)H_1(s), \tag{H:2}$$

$$X_{-1}(u)X_{-1}(v) = X_{-1}(u+v), \tag{Re:1a}$$

$$Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v), \tag{Re:1b}$$

$$Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}), \tag{Re:2}$$

$$\tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1}), \tag{Re:3}$$

$$\tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u), \quad \tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u), \tag{Re:4a,b}$$

$$\tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s), \quad \tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s), \tag{Re:5a,b}$$

$$H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su), \quad H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u), \tag{Re:6a,b}$$

$$H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u), \quad H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su). \tag{Re:6c,d}$$

For every $(\ell, j, k) \in E$:

$$X_{\ell,jk}(u)X_{\ell,jk}(v) = X_{\ell,jk}(u+v), \quad Y_{\ell,jk}(u)Y_{\ell,jk}(v) = Y_{\ell,jk}(u+v), \tag{Im:1a,b}$$

$$Y_{\ell,jk}(-t)X_{\ell,jk}(s)Y_{\ell,jk}(t) = X_{\ell,jk}\left(\frac{-t^{-1}}{c_{\ell j}}\right)Y_{\ell,jk}(-c_{\ell j}t^2s)X_{\ell,jk}\left(\frac{t^{-1}}{c_{\ell j}}\right), \tag{Im:2}$$

$$\tilde{w}_{\ell,jk}(s^{(\ell+1)(j-\ell)})\tilde{w}_{\ell,jk} = H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}), \tag{Im:3}$$

$$\tilde{w}_{\ell,jk}X_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = Y_{\ell,jk}\left(\frac{-u}{c_{\ell j}}\right), \quad \tilde{w}_{\ell,jk}Y_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = X_{\ell,jk}(-c_{\ell j}u), \tag{Im:4a,b}$$

$$\tilde{w}_{\ell,jk}H_1(s^{j-\ell})\tilde{w}_{\ell,jk}^{-1} = H_2(s^{-(\ell+1)}), \quad \tilde{w}_{\ell,jk}H_2(s^{\ell+1})\tilde{w}_{\ell,jk}^{-1} = H_1(s^{-(j-\ell)}), \tag{Im:5a,b}$$

$$H_1(s)X_{\ell,jk}(u)H_1(s)^{-1} = X_{\ell,jk}(s^{\ell+1}u), \quad H_2(s)X_{\ell,jk}(u)H_2(s)^{-1} = X_{\ell,jk}(s^{j-\ell}u), \tag{Im:6a,b}$$

$$H_1(s)Y_{\ell,jk}(u)H_1(s)^{-1} = Y_{\ell,jk}(s^{-(\ell+1)}u), \quad H_2(s)Y_{\ell,jk}(u)H_2(s)^{-1} = Y_{\ell,jk}(s^{-(j-\ell)}u). \tag{Im:6c,d}$$

Relations among the unipotent generators:

$$\begin{aligned}
(X_{-1}(u), X_{j-1,jk}(v)) &= 1, & (Y_{-1}(u), Y_{j-1,jk}(v)) &= 1 & \text{whenever } (j-1, j, k) \in E, & \quad (\text{U:1a,b}) \\
(Y_{-1}(u), X_{0,jk}(v)) &= 1, & (X_{-1}(u), Y_{0,jk}(v)) &= 1 & \text{whenever } (0, j, k) \in E, & \quad (\text{U:1c,d}) \\
(X_{\ell,jk}(u), Y_{m,pq}(v)) &= 1 & & & \text{if } j \neq p \text{ or } k \neq q \text{ or } |\ell - m| > 1, & \quad (\text{U:2}) \\
(X_{1,2k}(u), Y_{0,2k}(v)) &= X_{-1}(uv) & & & \text{whenever } (1, 2, k), (0, 2, k) \in E, & \quad (\text{U:3a}) \\
(X_{0,2k}(u), Y_{1,2k}(v)) &= Y_{-1}(uv) & & & \text{whenever } (0, 2, k), (1, 2, k) \in E, & \quad (\text{U:3b}) \\
\tilde{w}_{-1}X_{\ell,jk}(u)\tilde{w}_{-1}^{-1} &= X_{j-1-\ell,jk}((-1)^{j-\ell-1}u) & & & \text{whenever } (\ell, j, k), (j-1-\ell, j, k) \in E, & \quad (\text{U:4a}) \\
\tilde{w}_{-1}Y_{\ell,jk}(u)\tilde{w}_{-1}^{-1} &= Y_{j-1-\ell,jk}((-1)^{j-\ell-1}u) & & & \text{whenever } (\ell, j, k), (j-1-\ell, j, k) \in E. & \quad (\text{U:4b})
\end{aligned}$$

In (U:2) the triples (ℓ, j, k) and (m, p, q) range over E .

The group. Let $\mathcal{R} \subseteq F$ be the set of all relators above, and let $N_{\mathcal{R}}$ be its normal closure. Define

$$G = G_c = G(\mathfrak{m}) = F/N_{\mathcal{R}},$$

with quotient homomorphism $\pi: F \rightarrow G$. Write (G, G) for the commutator subgroup of G , i.e. the subgroup generated by all (g, h) with $g, h \in G$. It is a normal subgroup. From [Lemma 2.4](#) onward we use the same symbol for an element of F (a generator, $\tilde{w}_{-1}(s)$, $\tilde{w}_{\ell,jk}(s)$, ...) and for its image under π , whenever no confusion can arise.

Theorem 1.1. *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ satisfy $c(2) \geq 1$. Then the commutator subgroup of G_c is all of G_c :*

$$(G_c, G_c) = G_c.$$

The source statement concerns the specific c given by the coefficients of J . There $c(2) = 21493760 \geq 1$, so the source theorem is the special case of [Theorem 1.1](#) for that c . The formal theorem is stated for every c with $c(2) \geq 1$, which is exactly the property of c that the proof uses.

2 Lemmas

Throughout, $c: \mathbb{N} \rightarrow \mathbb{N}$ is arbitrary and $G = G_c$. The hypothesis $c(2) \geq 1$ is stated explicitly in each lemma where it is assumed.

2.1 Generalities on the presentation

Lemma 2.1 (Relators are trivial). *For every $r \in \mathcal{R}$ one has $\pi(r) = 1$ in G .*

Proof. We have $r \in \mathcal{R} \subseteq N_{\mathcal{R}} = \ker \pi$. □

Lemma 2.2 (Relations hold in G). *Let $a, b \in F$ be such that $ab^{-1} \in \mathcal{R}$. Then $\pi(a) = \pi(b)$.*

Proof. We have $ab^{-1} \in N_{\mathcal{R}} = \ker \pi$. Hence $\pi(a)\pi(b)^{-1} = 1$, that is, $\pi(a) = \pi(b)$. □

Lemma 2.3 (A subgroup containing the generators is everything). *Let $K \leq G$ be a subgroup with $\pi(x) \in K$ for every $x \in \mathcal{X}$. Then $K = G$.*

Proof. F is generated by $\mathcal{X}$ and π is surjective, so G is generated by $\pi(\mathcal{X})$. Since K contains $\pi(\mathcal{X})$, it contains the subgroup this set generates, namely G . $\square$

Lemma 2.4 (Images of commutators). *For all $a, b \in F$, $\pi((a, b)) \in (G, G)$.*

Proof. Since π is a homomorphism, $\pi(aba^{-1}b^{-1}) = (\pi(a), \pi(b))$. This is a commutator of two elements of G , hence lies in (G, G) . $\square$

2.2 Unipotent generators of the real root and the elements $\tilde{w}$

Lemma 2.5 ($X_{-1}(u)$ is a commutator). *Assume $c(2) \geq 1$. Then $X_{-1}(u) \in (G, G)$ for every $u \in \mathbb{C}$.*

Proof. Since $1 < 2$, $0 < 2$ and $1 \leq 1 \leq c(2)$, both $(1, 2, 1)$ and $(0, 2, 1)$ lie in E . Apply (U:3a) with $k = 1$ and parameters u and $v = 1$, using Lemma 2.2. This gives

$$X_{-1}(u) = X_{-1}(u \cdot 1) = (X_{1,21}(u), Y_{0,21}(1)) \quad \text{in } G.$$

The right-hand side is a commutator, hence in (G, G) . $\square$

Lemma 2.6 ($Y_{-1}(u)$ is a commutator). *Assume $c(2) \geq 1$. Then $Y_{-1}(u) \in (G, G)$ for every $u \in \mathbb{C}$.*

Proof. As before, $(0, 2, 1), (1, 2, 1) \in E$. By (U:3b) with $k = 1$, parameters u and $v = 1$, and Lemma 2.2, we get

$$Y_{-1}(u) = Y_{-1}(u \cdot 1) = (X_{0,21}(u), Y_{1,21}(1)).$$

This element lies in (G, G) by Lemma 2.4. $\square$

Lemma 2.7 ($\tilde{w}_{-1}(s)$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $\tilde{w}_{-1}(s) \in (G, G)$ for every $s \in \mathbb{C}^\times$.*

Proof. Since π is a homomorphism, the image of $\tilde{w}_{-1}(s)$ is $X_{-1}(s)Y_{-1}(-s^{-1})X_{-1}(s)$. Each factor lies in (G, G) by Lemmas 2.5 and 2.6. Since (G, G) is a subgroup, so does the product. $\square$

2.3 Unipotent generators of imaginary roots

Lemma 2.8 (A unit whose power is not 1). *For every integer $d > 0$ there exists $s \in \mathbb{C}^\times$ with $s^d \neq 1$.*

Proof. Take $s = 2$. If $2^d = 1$, then taking squared absolute values gives $4^d = |2^d|^2 = 1$. But $4^d > 1$ because $d \neq 0$, a contradiction. $\square$

Lemma 2.9 (Additivity of $X_{\ell,jk}$). *Let $(\ell, j, k) \in E$ and $u, v \in \mathbb{C}$. Then $X_{\ell,jk}(u+v) = X_{\ell,jk}(u)X_{\ell,jk}(v)$ in G .*

Proof. This is relation (Im:1a), via Lemma 2.2, together with $\pi(ab) = \pi(a)\pi(b)$. $\square$

Lemma 2.10 (Additivity of $Y_{\ell,jk}$). *Let $(\ell, j, k) \in E$ and $u, v \in \mathbb{C}$. Then $Y_{\ell,jk}(u+v) = Y_{\ell,jk}(u)Y_{\ell,jk}(v)$ in G .*

Proof. This is relation (Im:1b), via Lemma 2.2. $\square$

Lemma 2.11 (Differences for $X_{\ell,jk}$). *Let $(\ell, j, k) \in E$ and $a, b \in \mathbb{C}$. Then $X_{\ell,jk}(a - b) = X_{\ell,jk}(a)X_{\ell,jk}(b)^{-1}$ in G .*

Proof. Write $X = X_{\ell,jk}$.

1. By (Im:1a) with $u = v = 0$, $X(0)X(0) = X(0)$. The corresponding relator reduces in F to $X(0)$, so $X(0) = 1$ in G .
2. For $u \in \mathbb{C}$, Lemma 2.9 gives $X(u)X(-u) = X(u+(-u)) = X(0) = 1$. Hence $X(-u) = X(u)^{-1}$.
3. Finally, by Lemma 2.9,

$$X(a-b) = X(a+(-b)) = X(a)X(-b) = X(a)X(b)^{-1}. \quad \square$$

Lemma 2.12 (Differences for $Y_{\ell,jk}$). *Let $(\ell, j, k) \in E$ and $a, b \in \mathbb{C}$. Then $Y_{\ell,jk}(a-b) = Y_{\ell,jk}(a)Y_{\ell,jk}(b)^{-1}$ in G .*

Proof. Write $Y = Y_{\ell,jk}$. Relation (Im:1b) with arguments $a-b$ and b gives $Y(a-b)Y(b) = Y((a-b)+b) = Y(a)$. Therefore

$$Y(a-b) = (Y(a-b)Y(b))Y(b)^{-1} = Y(a)Y(b)^{-1}. \quad \square$$

Lemma 2.13 (A commutator from the torus action on $X_{\ell,jk}$). *Let $(\ell, j, k) \in E$, $s \in \mathbb{C}^\times$ and $v \in \mathbb{C}$. Then*

$$X_{\ell,jk}(s^{j-\ell}v)X_{\ell,jk}(v)^{-1} \in (G, G).$$

Proof. By (Im:6b) and Lemma 2.2, $X_{\ell,jk}(s^{j-\ell}v) = H_2(s)X_{\ell,jk}(v)H_2(s)^{-1}$ in G . Hence

$$X_{\ell,jk}(s^{j-\ell}v)X_{\ell,jk}(v)^{-1} = H_2(s)X_{\ell,jk}(v)H_2(s)^{-1}X_{\ell,jk}(v)^{-1} = (H_2(s), X_{\ell,jk}(v)),$$

which is a commutator. $\square$

Lemma 2.14 (A commutator from the torus action on $Y_{\ell,jk}$). *Let $(\ell, j, k) \in E$, $s \in \mathbb{C}^\times$ and $v \in \mathbb{C}$. Then*

$$Y_{\ell,jk}((s^{j-\ell})^{-1}v)Y_{\ell,jk}(v)^{-1} \in (G, G).$$

Proof. By (Im:6d) and Lemma 2.2, $Y_{\ell,jk}((s^{j-\ell})^{-1}v) = H_2(s)Y_{\ell,jk}(v)H_2(s)^{-1}$ in G . So the element in question equals $(H_2(s), Y_{\ell,jk}(v))$, a commutator. $\square$

Lemma 2.15 ($X_{\ell,jk}(u)$ lies in the commutator subgroup). *Let $(\ell, j, k) \in E$ and $u \in \mathbb{C}$. Then $X_{\ell,jk}(u) \in (G, G)$.*

Proof. Put $d = j - \ell$; then $d > 0$ because $\ell < j$.

1. By Lemma 2.8 choose $s \in \mathbb{C}^\times$ with $s^d \neq 1$.
2. Set $v = u/(s^d - 1)$. Then $u = (s^d - 1)v = s^d v - v$.
3. By Lemma 2.11,

$$X_{\ell,jk}(u) = X_{\ell,jk}(s^d v)X_{\ell,jk}(v)^{-1},$$

which lies in (G, G) by Lemma 2.13. $\square$

Lemma 2.16 ($Y_{\ell,jk}(u)$ lies in the commutator subgroup). *Let $(\ell, j, k) \in E$ and $u \in \mathbb{C}$. Then $Y_{\ell,jk}(u) \in (G, G)$.*

Proof. Put $d = j - \ell > 0$.

1. By Lemma 2.8 choose $s \in \mathbb{C}^\times$ with $s^d \neq 1$, and put $t = (s^d)^{-1}$. Then $t \neq 1$, since otherwise $s^d = 1$.

2. Set $v = u/(t-1)$. Then $tv - v = (t-1)v = u$.

3. By Lemma 2.12,

$$Y_{\ell,jk}(u) = Y_{\ell,jk}(tv - v) = Y_{\ell,jk}(tv) Y_{\ell,jk}(v)^{-1},$$

which lies in (G, G) by Lemma 2.14. $\square$

Lemma 2.17 ($\tilde{w}_{\ell,jk}(s)$ lies in the commutator subgroup). *Let $(\ell, j, k) \in E$ and $s \in \mathbb{C}^\times$. Then $\tilde{w}_{\ell,jk}(s) \in (G, G)$. No hypothesis on $c(2)$ is needed.*

Proof. In G we have

$$\tilde{w}_{\ell,jk}(s) = X_{\ell,jk}(s) Y_{\ell,jk}(-s^{-1}/c_{\ell j}) X_{\ell,jk}(s).$$

By Lemmas 2.15 and 2.16, each factor lies in (G, G) . Hence the product lies in (G, G) . $\square$

2.4 The torus generators

Lemma 2.18 ($H_1(-s)H_2(-s^{-1})$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $H_1(-s)H_2(-s^{-1}) \in (G, G)$ for every $s \in \mathbb{C}^\times$.*

Proof. By (Re:3) and Lemma 2.2, $H_1(-s)H_2(-s^{-1}) = \tilde{w}_{-1}(s)\tilde{w}_{-1}(1)$ in G . Both factors lie in (G, G) by Lemma 2.7. $\square$

Lemma 2.19 ($H_1(x)H_2(x^{-1})$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $H_1(x)H_2(x^{-1}) \in (G, G)$ for every $x \in \mathbb{C}^\times$.*

Proof. Apply (Re:3) with $s = -x$. Since $-(-x) = x$ and $-((-x)^{-1}) = x^{-1}$, it reads $\tilde{w}_{-1}(-x)\tilde{w}_{-1}(1) = H_1(x)H_2(x^{-1})$. By Lemma 2.2 this holds in G . The left side lies in (G, G) by Lemma 2.7. $\square$

Lemma 2.20 (Consequence of (Im:3)). *Let $(\ell, j, k) \in E$ and $s \in \mathbb{C}^\times$. Then $H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}) \in (G, G)$.*

Proof. By (Im:3) and Lemma 2.2,

$$H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}) = \tilde{w}_{\ell,jk}(s^{(\ell+1)(j-\ell)})\tilde{w}_{\ell,jk}(1)$$

in G . Both factors lie in (G, G) by Lemma 2.17. $\square$

Lemma 2.21 ($H_1(y^2)H_2(y)$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $H_1(y^2)H_2(y) \in (G, G)$ for every $y \in \mathbb{C}^\times$.*

Proof. Since $0 < 2$ and $1 \leq c(2)$, we have $(0, 2, 1) \in E$, with $j - \ell = 2$ and $\ell + 1 = 1$. Relation (Im:3) for this triple and $s = -y$ reads

$$\tilde{w}_{0,21}((-y)^2)\tilde{w}_{0,21}(1) = H_1(y^2)H_2(y),$$

using $-(-y)^2 = y^2$ and $-(-y)^1 = y$. Since $(-y)^2 = y^2$, the left side is $\tilde{w}_{0,21}(y^2)\tilde{w}_{0,21}(1)$. By Lemma 2.2 the identity holds in G . The left side lies in (G, G) by Lemma 2.17. $\square$

Lemma 2.22 ($H_1(z^3)$ as a quotient). *For every $z \in \mathbb{C}^\times$, the following holds in G (no hypothesis on c):*

$$H_1(z^3) = (H_1(z)H_2(z^{-1}))(H_1((z^{-1})^2)H_2(z^{-1}))^{-1}.$$

Proof. First we show that $s \mapsto H_1(s)$ is a group homomorphism $\mathbb{C}^\times \rightarrow G$.

1. By (H:1a) and [Lemma 2.2](#), $H_1(st) = H_1(s)H_1(t)$ in G for all $s, t \in \mathbb{C}^\times$.
2. Taking $s = t = 1$ gives $H_1(1)H_1(1) = H_1(1)$, hence $H_1(1) = 1$.
3. Taking $t = s^{-1}$ gives $H_1(s)H_1(s^{-1}) = H_1(1) = 1$, hence $H_1(s^{-1}) = H_1(s)^{-1}$.
4. By induction on n , $H_1(x^n) = H_1(x)^n$ for all $n \in \mathbb{N}$.

In particular $H_1((z^{-1})^2) = H_1(z)^{-2}$. Therefore the right-hand side equals

$$H_1(z) H_2(z^{-1}) H_2(z^{-1})^{-1} H_1(z)^2 = H_1(z)^3 = H_1(z^3). \quad \square$$

Lemma 2.23 (Cube roots in $\mathbb{C}^\times$). *For every $s \in \mathbb{C}^\times$ there is $z \in \mathbb{C}^\times$ with $z^3 = s$.*

Proof. Since $\mathbb{C}$ is algebraically closed, there is $w \in \mathbb{C}$ with $w^3 = s$. Then $w \neq 0$, because otherwise $s = 0^3 = 0$. Take $z = w$. $\square$

Lemma 2.24 ($H_1(s)$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $H_1(s) \in (G, G)$ for every $s \in \mathbb{C}^\times$.*

Proof. 1. By [Lemma 2.23](#) write $s = z^3$ with $z \in \mathbb{C}^\times$.

2. By [Lemma 2.19](#), $H_1(z)H_2(z^{-1}) \in (G, G)$.
3. By [Lemma 2.21](#) with $y = z^{-1}$, $H_1((z^{-1})^2)H_2(z^{-1}) \in (G, G)$.
4. Since (G, G) is a subgroup, the product of the first element with the inverse of the second lies in (G, G) . By [Lemma 2.22](#) this product is $H_1(z^3) = H_1(s)$. $\square$

Lemma 2.25 ($H_2(s)$ as a conjugate of $H_1(s)$). *For every $s \in \mathbb{C}^\times$ one has $H_2(s) = \tilde{w}_{-1} H_1(s) \tilde{w}_{-1}^{-1}$ in G .*

Proof. This is relation (Re:5a), via [Lemma 2.2](#). $\square$

Lemma 2.26 ($H_2(s)$ lies in the commutator subgroup). *Assume $c(2) \geq 1$. Then $H_2(s) \in (G, G)$ for every $s \in \mathbb{C}^\times$.*

Proof. By [Lemma 2.25](#), $H_2(s)$ is a conjugate of $H_1(s)$, and $H_1(s) \in (G, G)$ by [Lemma 2.24](#). Since (G, G) is normal in G , it contains every conjugate of its elements. (That $\tilde{w}_{-1}$ itself lies in (G, G) , by [Lemma 2.7](#), is not needed.) $\square$

3 Proof of the theorem

Proof of Theorem 1.1. Assume $c(2) \geq 1$. By [Lemma 2.3](#) it suffices to show that the image of every symbol in $\mathcal{X}$ lies in (G, G) . We distinguish the six kinds of symbols:

- $H_1(s)$, $s \in \mathbb{C}^\times$: [Lemma 2.24](#);
- $H_2(s)$, $s \in \mathbb{C}^\times$: [Lemma 2.26](#);
- $X_{-1}(u)$, $u \in \mathbb{C}$: [Lemma 2.5](#);
- $Y_{-1}(u)$, $u \in \mathbb{C}$: [Lemma 2.6](#);
- $X_{\ell,jk}(u)$, $(\ell, j, k) \in E$, $u \in \mathbb{C}$: [Lemma 2.15](#);
- $Y_{\ell,jk}(u)$, $(\ell, j, k) \in E$, $u \in \mathbb{C}$: [Lemma 2.16](#).

Hence $(G, G) = G$. $\square$

Relation to the source proof. The argument follows the source proof step by step: first X_{-1} and Y_{-1} via (U:3a,b); then $X_{\ell,jk}$ and $Y_{\ell,jk}$ via (Im:6b,d); then the elements $\tilde{w}$; then H_1 via (Re:3), (Im:3) for the triple $(0, 2, k)$, and cube roots; finally H_2 via (Re:5a). The deviations are minor.

- *The parameter c .* The coefficient function c is an arbitrary parameter, subject only to $c(2) \geq 1$, instead of the coefficients of J .
- *Concrete choices.* For X_{-1} and Y_{-1} the proof takes $k = 1$ and $v = 1$, writing $z = z \cdot 1$. This also bypasses a misprint in the source's displayed formula for Y_{-1} . The unit with $s^{j-\ell} \neq 1$ is $s = 2$.
- *Elementary facts made explicit.* The facts used implicitly in the source are proved: $X_{\ell,jk}(0) = 1$, $X_{\ell,jk}(-u) = X_{\ell,jk}(u)^{-1}$, and that $s \mapsto H_1(s)$ is a homomorphism (Lemmas 2.11, 2.12 and 2.22).
- *The step for H_2 .* It uses only normality of the commutator subgroup, not the membership $\tilde{w}_{-1} \in (G, G)$ invoked in the source.
- *Relations used.* The proof uses only (H:1a), (Re:3), (Re:5a), (Im:1a,b), (Im:3), (Im:6b,d) and (U:3a,b).
- *Unused lemmas.* Lemmas 2.1, 2.10, 2.18 and 2.20 are proved but not needed for the main theorem.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.1 is `gsimple_g_m_perfect`.

Lemma	Lean declaration	Status
Lemma 2.1	<code>mk_rel_one</code>	verified
Lemma 2.2	<code>mk_eqr</code>	verified
Lemma 2.3	<code>subgroup_top_of_gens</code>	verified
Lemma 2.4	<code>cm_mem_commutator</code>	verified
Lemma 2.5	<code>xm_mem</code>	verified
Lemma 2.6	<code>ym_mem</code>	verified
Lemma 2.7	<code>wm_mem</code>	verified
Lemma 2.17	<code>ww_mem</code>	verified
Lemma 2.8	<code>exists_unit_pow_ne_one</code>	verified
Lemma 2.9	<code>xx_add_mk</code>	verified
Lemma 2.10	<code>yy_add_mk</code>	verified
Lemma 2.11	<code>xx_sub_mk</code>	verified
Lemma 2.12	<code>yy_sub_mk</code>	verified
Lemma 2.13	<code>xx_conj_mem</code>	verified
Lemma 2.14	<code>yy_conj_mem</code>	verified
Lemma 2.15	<code>xx_mem</code>	verified
Lemma 2.16	<code>yy_mem</code>	verified
Lemma 2.18	<code>h1h2_neg_mem</code>	verified
Lemma 2.19	<code>h1h2_inv_mem</code>	verified
Lemma 2.20	<code>im3_mem</code>	verified
Lemma 2.21	<code>h1sq_h2_mem</code>	verified

Lemma	Lean declaration	Status
Lemma 2.22	<code>h1_cube_eq</code>	verified
Lemma 2.23	<code>exists_unit_cube</code>	verified
Lemma 2.24	<code>h1_mem</code>	verified
Lemma 2.25	<code>h2_eq_conj</code>	verified
Lemma 2.26	<code>h2_mem</code>	verified